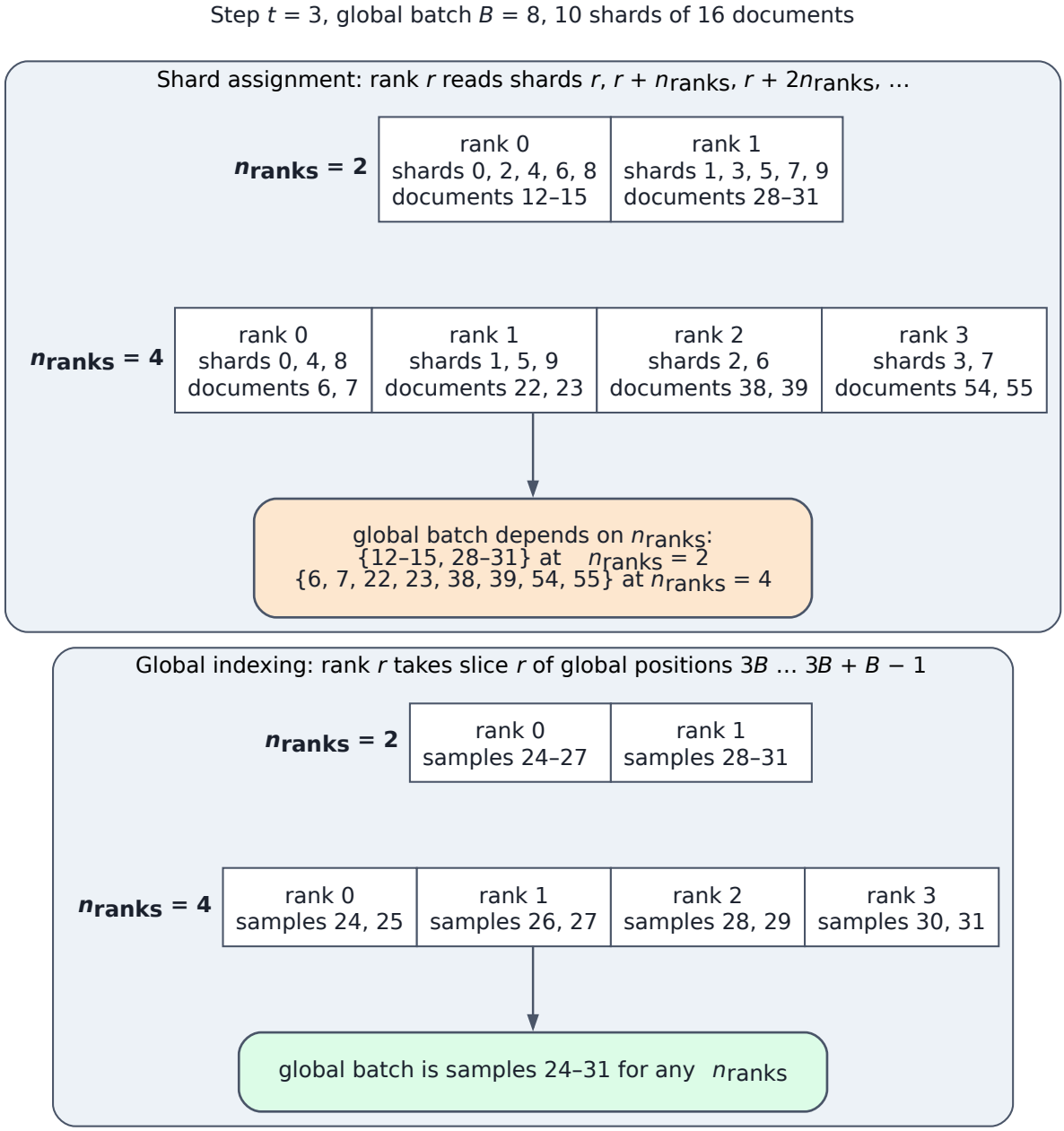
Step $t = 3$, global batch $B = 8$, 10 shards of 16 documents
Shard assignment: rank r reads shards r, $r + n_{\text{ranks}}$, $r + 2n_{\text{ranks}}$, ...
$n_{\text{ranks}} = 2$
rank 0
shards 0, 2, 4, 6, 8
documents 12–15
rank 1
shards 1, 3, 5, 7, 9
documents 28–31
$n_{\text{ranks}} = 4$
rank 0
shards 0, 4, 8
documents 6, 7
rank 1
shards 1, 5, 9
documents 22, 23
rank 2
shards 2, 6
documents 38, 39
rank 3
shards 3, 7
documents 54, 55
global batch depends on n_{ranks}:
{12–15, 28–31} at $n_{\text{ranks}} = 2$
{6, 7, 22, 23, 38, 39, 54, 55} at $n_{\text{ranks}} = 4$
Global indexing: rank r takes slice r of global positions $3B$ ... $3B + B - 1$
$n_{\text{ranks}} = 2$
rank 0
samples 24–27
rank 1
samples 28–31
$n_{\text{ranks}} = 4$
rank 0
samples 24, 25
rank 1
samples 26, 27
rank 2
samples 28, 29
rank 3
samples 30, 31
global batch is samples 24–31 for any n_{ranks}